COMPUTER SCIENCE TRIPOS Part IB – 2024 – Paper 5

1 Computer Networking (awm22)

http v1.2
TCP
IPv4
IEEE 802.11 (WiFi)
2.4 GHz wireless band

Consider this communications-stack diagram. Each layer of the stack can provide multiplexing support to the layer above, mapping a single channel into multiple independent communications channels.

For each of the following examples of the diagram's communications-stack layers, discuss how multiplexing is put into practice, including an example of how relevant data is identified and how such data is mapped to and from the layer above.

Answer:

The question is intended to encourage students to differentiate the different multiplexing possible at different communications layers.

broadly speaking marking schedule is
1 marks to make clear the entities
1 mark to make clear the nature of the multiplexing
1 mark to identify how a specific communications flow is identified
1 mark to note how (non relevant) data is ignored or special circumstance where data is not ignored.

(*a*) A 2.4 GHz physical layer. [4 marks]

Answer:

The 802.11 WiFi system divides up the allocated physical band into channels; some (although not all) channels can operate simultaneously without interference.

Multiplexing in this context transforms a large allocated bandwidth into smaller bandwidth channels.

Each transmitter and receiver are tuned (analog transmitter/receiver) to a specific channel and will not decode other channels, perceiving them only as signal noise - if at all.

(*b*) A data-link layer. [4 marks]

Answer:

Addresses permit an interface to uniquely identify packets destined for itself. Additionally addresses (eg the 48-bit Ethernet addresses) allow a sender to identify itself to a receiver.

Each interface may receive at least one unique address (along with packets addressed with a number of special-purpose broadcast or multicast destinations.)

A receiver interface, by default, will ignore packets not intended for it.

(*c*) An IPv4 network layer. [4 marks]

Answer: addresses identify destinations for each packet.

Multiple packets of any (combination of sender and) receiver can exist in the data-link layer but there is a mapping from each data-link layer to a single IP address.

Multiplexing into a specific entity (interface) requires the identification of a unique (physical-layer) address and maps this to one or more IP addresses.

Each host interface will associate to a specific IP Address.

These in turn are decoded to "proto" or "service", TCP and UDP layers in the transport

Because each IP packet is entirely independent; this is perhaps the easiest multiplexing layer. Each packet is directed to (eg. UDP, or TCP) processing based entirely on the content of the proto field.

(*d*) A TCP transport layer. [4 marks]

Answer:

once each IP address reaches the appropriate interface, the header is parsed, a field within TCP header nominates a destination port number to pass th epacket - this port identifies the local (TCP) endpoint.

There is a more sophisticated point to be made here; the individual communications flow is identified by the tuple consisting the address and port of the client and the address and port of the server

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(*e*) A web server using HTTP v1.2. [4 marks]

Answer:

HTTP requires each request to specify the name of the host from which a retrieval will take place - this permits a single server to serve content for multiple web servers. If the server entry is not specified or specifies a server incorrectly, an error is returned. In this way the URL alone is not the canonical identifier, the identifier becomes a combination of URL and the machine from which the request is made.
